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Salt-in-Polymer Electrolytes Based on Polysiloxanes for Lithium-Ion Cells: Ionic Transport and Electrochemical Stability

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In this work, the ion transport and the electrochemical stability has been investigated for cross linked salt-in-polysiloxane membranes containing the two lithium salts with boron based anions, i.e. LiDFOB (lithium difluoro(oxalato)borate) and LiBOB (lithium bis(oxalato)borate) have been investigated. The stability was characterized by cyclic voltammetry at 70 °C in a three electrode cell with Li as counter and reference electrodes and a nickel or platinum working electrode. The electrochemical stability window of polysiloxane based electrolytes ranged between 0 V and 4.7 V vs. Li/Li^+. The high ionic conductivity of 6.61 x 10^{-2} mS/cm at 30 °C has been examined with impedance spectroscopy on blocking electrodes. The investigated polymer electrolytes showed an outstanding thermal resistance as confirmed by thermal analysis with DSC. Furthermore, analysis of the lithium transference number has been carried out using the potentiostatic polarization method introduced by BRUCE and VINCENT.

Introduction

The invention and commercialization of lithium ion batteries has established a new technology branch on the battery market (1). Some of the commercially available cells contain a swollen polymer electrolyte which consists of PVdF-hexafluoropropylene as a matrix and a liquid electrolyte like $LiPF_6$/EC/DMC as the plasticizer (2). For the successful introduction of new polymer matrices, a detailed investigation of the basic polymer system is essential. The amount of work done on polymer electrolytes for lithium ion batteries (LIB) is enormous. The materials can be separated into two major categories, solid polymer electrolytes (SPE) and gel polymer electrolytes (GPE). The investigations of FENTON et *al*. depict the first solid polymer electrolytes with the use of poly(ethylene oxide (PEO) and a conducting salt (3). Afterwards in 1979 ARMAND *et al.* proposed that polymer electrolytes based on PEO with incorporated lithium salts could be used for LIB (4). These SPEs have been extensively investigated during the last thirty years (5-7). In the literature there are several polymers which are derived from PEO, i.e. the polyphosphazene MEEP and the polysiloxanes with oligoether groups (8-13). They combine a good lithium salt solubility and are amorphous even at room temperature due to high flexibility of the main chain.

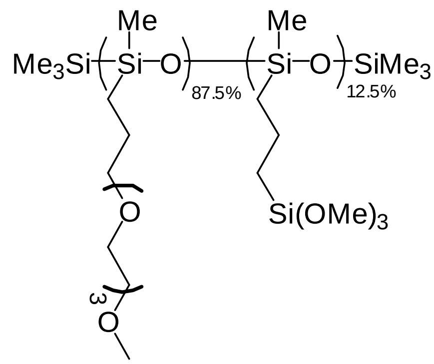

Figure 1. Polysiloxane PSx-[P(EO)$_3$OMe]$_{87.5\%}$[PSi(OMe)$_3$)]$_{12.5\%}$ with 12.5 wt.-% Si(OMe)$_3$ linking groups and 87.5 wt.-% oligoether, linking leads to formation of Si-O-Si bonds between the Si(OMe)$_3$ groups (13).

In the last ten years there has been a clear progress with respect to higher total conductivities, higher transference numbers and an appropriate electrochemical stability. Current research publications in this area present polysiloxane liquids containing oligethylene oxide groups which are electrochemically stable up to 4.4 V vs. Li/Li$^+$ (14). WALKOWIAK *et al.* showed that the use of tripodant side-chains leads to good total ionic conductivities (10^{-3} S/cm at 25 °C) (15). Recently, it was also reported that cross-linked cyclosiloxanes with oligoethylene oxide **s**howed oxidation stability up to 4.8 V vs. Li/Li$^+$ (16). WANG *et al.* presented the synthesis and analysis of polysiloxane based electrolytes with poly(ethyl glycol) methyl ether methacrylate side-chains. These polymers had ionic conductivities of 1.15 x 10^{-4} S/cm at 25 °C with lithium transference numbers varying from 0.26 to 0.33 (17-18). ZHANG *et al*. presented a polysiloxane with a total conductivity of 1.33 x 10^{-4} S/cm at 25 °C (19). SNYDER *et al.* synthesized and characterized a polyelectrolyte with fixed anion sites and mobile lithium ions which exhibits a transference number of t_+=1, but at the cost of a relatively low overall conductivity of 2.5 x 10^{-6} S/cm (20). The synthesis of polymer electrolytes has been optimized to create reproducible, free standing membranes (13). In this work, two promising lithium salts have been tested in polysiloxane membranes which can be used as an alternative to $LiPF_6$. SPEs made of polysiloxane and LiBOB and their film forming abilities have been intensively investigated by NAKAHARA *et al.* in 2006 (21-24). Except some patents there are no publications available for polysiloxane membranes containing LiDFOB.

Therefore the electrochemical characterization including electrochemical stability window and temperature dependent transference numbers of a selected polysiloxane system containing 12.5 wt.-% Si(OMe)$_3$ cross-linking groups and 87.5 wt.-% oligoether groups (PSx 12.5) is presented (Figure 1).

The polymer electrolyte membranes are intended to be used in safe batteries that can withstand elevated temperatures (up to 70 °C or higher). Thus the SPE shown has to be resistant to high potentials (U_{min}~4.0 V vs. Li/Li$^+$) when used in lithium ion batteries at these temperatures. The electrochemical stability window was examined by cyclic voltammetry (CV) at 70 °C. The electrochemical stability (vs. Li/Li$^+$) of the salts studied in this paper were reported in the literature for different solvents as follows: LiDFOB: 4.5 V (determined on platinum in EC:PC:EMC = 1:1:3, 0.1 mV s-1) (25) and LiBOB: 4.5 V (determined on platinum in PC, 1.0 mV s^{-1}) (26). Investigations by ZUGMANN et al. concerning hydrolysis of LiDFOB in water showed no HF-formation (27). This is in line with the investigations of FU *et al.* who figured out that the use of LiDFOB-electrolytes with $LiMn_2O_4$-cathodes causes no Mn-dissolution which is the case for $LiPF_6$ based electrolytes (25). The salt LiDFOB is also discussed as a functional additive for lithium ion batteries (28).

Experimental

Synthesis of polysiloxane PSx-$[P(EO)_3OMe]_{87.5\%}[PSi(OMe)_3)]_{12.5\%}$

The polysiloxane PSx-$[P(EO)_3OMe]_{87.5\%}[PSi(OMe)_3]_{12.5\%}$ (PSx 12.5) can be synthesized by the following procedure. The basic polymer poly(methylhydro)siloxane (PMHS, averaging 35 repeating units, 1700-3200 g/mol) was purchased from *Sigma Aldrich* and was stored over molecular sieve (4 Å, Fluka). Toluene and pentane were purchased from *VWR* and dried according to standard procedure over sodium/benzophenone. Allyltrimethoxysilane (97%) and platinum-divinyltetramethyldisiloxane complex in xylene (2.1 – 2.4% Pt, Karstedt's catalyst) were purchased from *ABCR* and used as received.

In a dry 250 mL SCHLENK flask, 6.5 g (0.10 mol) of poly(methylhydro)siloxane were stored under inert atmosphere. The polymer was degassed for about 30 minutes under low pressure vacuum ($2 \cdot 10^{-3}$ mbar) and vigorous stirring. Subsequently, 100 mL of anhydrous toluene and 2.03 g (0.0125 mol) allyltrimethoxysilane were added. To start the reaction, 100 µL of Karstedt's catalyst (Platinum-divinyltetramethyldisiloxane complex in xylene, 2.1 – 2.4% Pt) were added. The reaction mixture was heated to 70 °C until no remaining allyl protons could be detected in IR (ν = 1634 cm^{-1}, usually between 2 and 4 days). Afterwards, 20.43 g (0.10 mol) of triethylene glycol allyl methyl ether were added in order to replace the residual protons. When Si-H protons could no longer be detected by IR and ^{1}H-NMR (usually between 10 and 14 days), the platinum catalyst was inactivated and removed by charging the solution with activated charcoal. To remove the charcoal, the reaction solution was filtered over *Celite*® under inert gas atmosphere. The solvent was evaporated under low pressure vacuum ($2 \cdot 10^{-3}$ mbar) the polymer was washed with pentane in order to remove the excess of triethylene glycol side chain.

^{1}H-NMR ($CDCl_3$, ppm, δ): 0.04, 0.45, 0.55, 0.64, 1.55, 3.33, 3.50-3.62,

^{13}C-NMR ($CDCl_3$, ppm, δ): -0.51, 13.32, 16.38, 21.59, 50.53, 58.32, 68.19-71.67.

Preparation of polysiloxane membranes

LiBOB (lithium bis(oxalato)borate) (*Chemetall*, battery grade) was dried under high vacuum (10^{-6} mbar) for 5 days at 60 °C and dissolved in THF. LiDFOB (lithium difluoro(oxalato)borate) was synthesized by SCHREINER *et al.* (29) and was used as a solution in THF. In a preparation flask, approximately 0.75 g of the polysiloxane PSx-$[P(EO)_3OMe]_{87.5\%}[PSi(OMe)_3)]_{12.5\%}$ and 0.5 mL of dry THF were stirred well before adding the necessary amount of salt (LiDFOB or LiBOB, stock solution in THF, c = 0.1 – 0.3 g/mL). The salt concentration was taken as 0.61 mmol per gram of polymer. To start cross-linking, 75 µL methanolic hydrochloride solution (c = 1.25 M) were added and the reaction solution was stirred for 6 h until a slight increase in viscosity was observed. The solution was cast into a PTFE dish (Ø=4 cm) and the dish was stored in a pre-heated reaction vessel (70 °C) over night under nitrogen gas atmosphere. Subsequently the solvent was evaporated under reduced pressure (100 mbar) at 70 °C. To dry the membrane completely, the mixture was stored under these conditions for about 7 to 10 days. The resulting transparent and free-standing membranes had a thickness between 100 and 300 µm and an average water content of 12.6 ppm (determined by the Karl-Fischer-Titration).

Instrumental

All cells were thermo stated (Phoenix II, Haake). The operation temperature was stable within ± 0.1 K. Cyclic voltammetry was performed using a computer controlled potentiostat (Autolab PGSTAT302N, Metrohm). Ionic conductivities of the polymer electrolyte membranes were obtained by impedance spectroscopy using a Solatron SI-1260 impedance analyzer and an EG&G potentiostat (model 283) in the frequency range between 0.1 Hz and 100 kHz with an AC amplitude of 40 mV.

Cell assembly

The electrochemical stability window (ESW) measurements were carried out in modified *Swagelok*-T-cells assembled under Argon atmosphere with controlled moisture (<0.6 ppm) and oxygen (<0.1 ppm) levels (30). Lithium foils (*Chemetall*, 99.99%) were used as reference (RE: Ø=5 mm)- and counter electrodes (CE: Ø=12 mm). For anodic and cathodic decomposition platinum (*Chempur*, 99.99%, Ø =2 mm) and nickel (*Alfa Aesar*, 99.98%,) were chosen as polarizable working electrodes (WE). Before every measurement, the working electrodes were manually polished with abrasive paper up to grain P4000. The investigated membranes were cut into circular discs (Ø=12 mm) with a mean thickness of 250 µm. All impedance and transference number measurements were performed by using custom-made stainless steel measurement cells, the temperature range was between 30 °C and 70 °C for impedance measurements in the context of conductivity determinations and between 50 °C and 100 °C for transference experiments.

Evaluation of the transference experiments

The lithium transference number (t_+) was determined with the method introduced by BRUCE and VINCENT (31). The typical experiment consists of an ac impedance measurement combined with a dc polarization technique in a symmetric cell (Li|Polymer|Li). A two electrode *Swagelok*-cell was used (A = 1.13 cm²). The impedance was measured between 200 kHz and 1 Hz with an ac amplitude of 10 mV. The impedance contribution of the Li|polymer interface (due to passivating layers) was determined with ac impedance both, before and at the end of a steady-state dc polarization. The analysis of the frequency dependent impedance was analyzed and the corresponding charge-transfer resistances $R_{0,b}$ (homogeneous sample before polarization) and $R_{SS,b}$ (polarized sample after reaching the steady-state) were determined.

For the dc polarization, the applied voltage ΔV was set to 10 mV. The time dependence of the cell current was monitored and the initial current (I_0) after switching-on the dc voltade and the steady-state current (I_{SS}) were taken for the final evaluation.

Under these conditions, the calculation of t_+ was typically carried out using the following equation:

$$t_+ = \frac{I_{SS}\left(\Delta V - I_0 R_{0,b}\right)}{I_0\left(\Delta V - I_{SS} R_{SS,b}\right)} \quad [1]$$

An additional correction may be introduced according to ABRAHAM *et al.* due to the fact that the polymer electrolyte resistance changes during the polarization from $R_{ely,0}$ to $R_{ely,ss}$ which leads to the following equation (32):

$$t_+ = \frac{I_{SS} R_{ely,SS}\left(\Delta V - I_0 R_{0,b}\right)}{I_0 R_{ely,0}\left(\Delta V - I_{SS} R_{SS,b}\right)} \quad [2]$$

Evaluation of the electrochemical stability

The evaluation of the ESW is usually based upon linear sweep voltammetry in three-electrode cells with an inert working electrode. In the literature, the cutoff criterion for the starting decomposition was most often taken as the voltage where the exponentially increasing current reached values of 1 mA/cm² (33-34). For polymer electrolytes, this analysis is often done by cyclic voltammetry without a concrete cutoff current density (14, 17, 32). Although values for the anodic oxidation potentials of polyethylene oxide-type polymers were reported as low as 4.0 V (2), most authors agree upon higher anodic stability limits up to 4.5 V (15, 17, 32, 35-36). Normally, the observed starting current densities in the oxidation kinetics of polymer electrolytes are rather small. This may be an advantage as compared to liquid electrolytes.

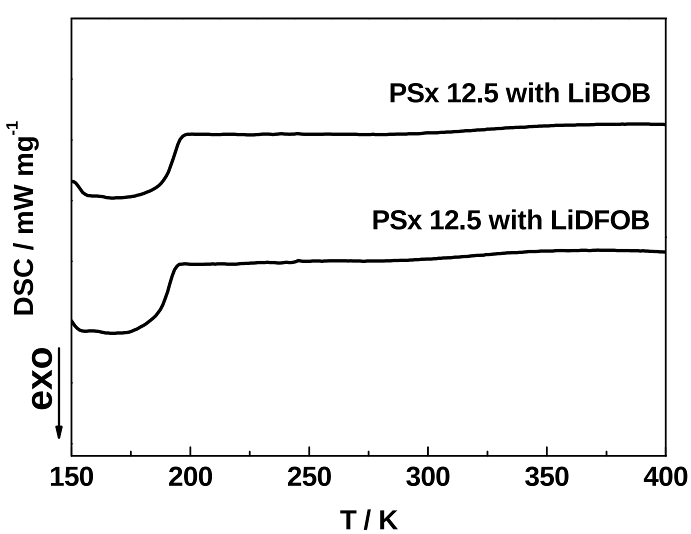

Figure 2. DSC heating curves for the PSx 12.5 polymer electrolyte membranes containing the two investigated lithium salts.

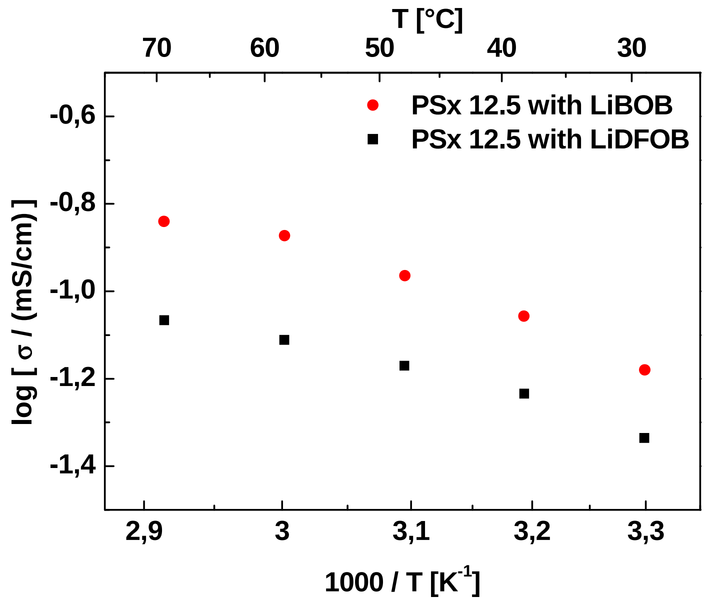

Figure 3. Total ionic conductivity for PSx 12.5 of the investigated lithium salts.

For the determination of the stable potential range of the polymer electrolytes, cyclic voltammetry (CV) was used with a scan rate of 1 mV/s. Anodic and cathodic potential limits were determined with two different working electrode materials in the same cell. The potential values were taken at a cutoff current density of 0.05 mA/cm². This cutoff criterion for anodic decomposition is low enough to prevent overly optimistic values removed from application, yet high enough to distinguish between the superimposed double layer charging current and currents induced by oxidation of impurities from faradaic oxidation currents of the polymer electrolyte or the salt.

Thermal Properties of PSx 12.5

DSC (differential scanning calorimetry) was used to analyze the thermal properties of the polymer electrolyte PSx-$[P(EO)_3OMe]_{87.5\%}[PSi(OMe)_3)]_{12.5\%}$ (PSx 12.5) containing different salts. Figure 2 shows the heating curves of polymer electrolyte membranes measured in the temperature range between 150 K and 425 K. The membranes show crystallinity in neither the main nor the side chains. The glass transition temperatures (T_g) were determined at the onset of the inflection in the heating curve (ca. 200 K). Thermal decomposition could not be detected in the observed temperature range which is in line with results of KANG *et al.* (35).

Results and Discussion

Total Ionic Conductivity

The total ionic conductivities of salt-in-polymer electrolyte membranes made of PSx 12.5 were measured. Two sample series with salt concentrations of 0.61 mmol/$g_{polymer}$ LiDFOB (lithium bis(fluoro dioxalato)borate) and 0.61 mmol/$g_{polymer}$ LiBOB (lithium bis(oxalato borate)) were compared. Figure 3 shows the ionic conductivities as a function of temperature.

It is obvious, that polymer electrolyte membranes containing LiBOB (4.57×10^{-2} mS/cm at 30 °C) show higher ionic conductivities than polymer electrolyte membranes containing comparable amounts of LiDFOB (6.61×10^{-2} mS/cm at 30 °C). It may be explained by higher dissociation of LiBOB due to the highly symmetric BOB^- anion.

Electrochemical Stability of PSx-$Si(OMe)_3$ with LiDFOB and LiBOB

Figs. 4 and 5 show cyclic voltammetry (CV) curves at 70 °C for the two types of polysiloxane based membranes. They contain separate voltage scans in the high and the low potential range, each of which starts at a medium potential near 3 V (vs. Li/Li^+). The CV curve of the polysiloxane with 0.61 mmol/$g_{polymer}$ LiDFOB in Fig. 4 shows anodic oxidation starting around 4.7 V vs. Li/Li^+ due to the oxidation of the oligoether side groups of the polymer and also due to the lithium salt, as confirmed by several other works for similar polymers with ether groups (17, 25-26, 32, 35, 37). The steep increase of the current density at 5.3 V vs. Li/Li^+ is assigned to increasing anion oxidation.

The cathodic region of the CV on nickel electrodes showed a huge plating/stripping peak around 0 V and three minor reduction peaks at 1.3 - 1.8, 0.8 - 0.9 and 0.4 - 0.5 V vs. Li/Li^+. The current densities of these three peaks are in the range of 0.01-0.02 mA/cm² and therefore too small to be visible in Fig. 4. The peaks at 0.5 and 0.8 V have also been observed by other authors investigating underpotential deposition of lithium on nickel surfaces (38). A very small, but broad peak in the range 1.3 - 1.8 V is most probably due to slow reduction of the anion of LiDFOB, as a similar observation has been made for LiBOB at lithium electrodes and for potentials around 1.75 V (39). The coulombic efficiency of the plating/stripping process (62%) showed that a slow side reaction occurs between the polymer and fresh metallic lithium which gives rise to a interface layer with considerable impedance as discussed below.

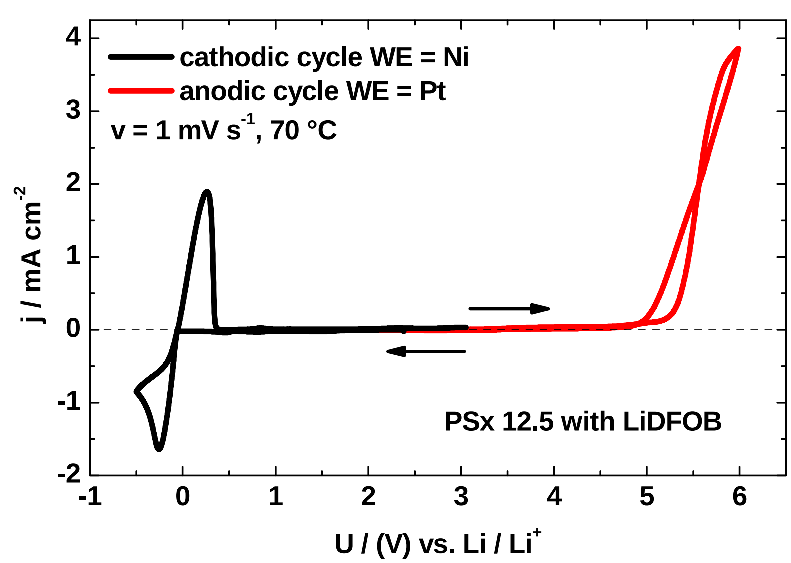

Figure 4: Cyclic voltammetry of PSx 12.5 containing 0.61 mmol/$g_{polymer}$ LiDFOB.

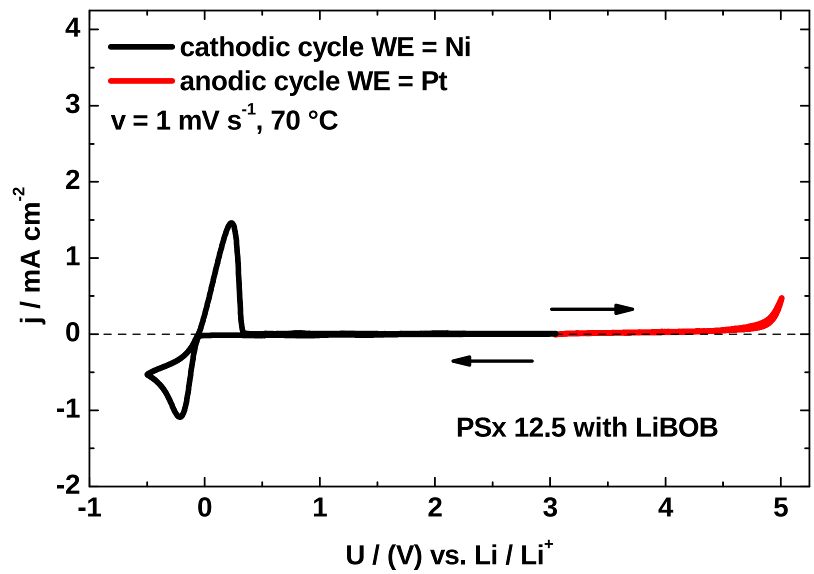

Figure 5: Cyclic voltammetry of PSx 12.5 containing 0.61 mmol/$g_{polymer}$ LiBOB.

The CV curves of the polymer electrolyte made of polysiloxane with 0.61 mmol/$g_{polymer}$ LiBOB (Figure 5) showed a similar behavior as with LiDFOB. A slight difference is that a clear oxidation starts at a lower potential (at 5.0 V) as compared to 5.3 V for PSx 12.5 with LiDFOB. A very slow decomposition current due to the oligoether side chains was observed before starting at a lower potential, i.e. 4.5 V.

The cathodic processes in the low potential range are very similar between Figs. 4 and 5. The lithium plating/stripping process in Fig. 5 for the LiBOB containing electrolyte showed a coulombic efficiency of 68%.

Lithium Transference Numbers

The transference number of lithium ions is a central property besides the total conductivity to judge a lithium battery electrolyte. The transference number of lithium ions describes the fraction of the total conductivity which is available for lithium ion transport. Accordingly, it can be used to characterize the maximum rate due to lithium ion transport under steady state conditions in lithium ion cells. The total conductivity of our polysiloxane based electrolytes was shown in Fig. 3 for the relevant temperature range, as measured by standard impedance spectroscopy (1 – 20 kHz). In conventional liquid electrolytes, the lithium transference numbers reach values near or slightly lower than 0.5. In polymer electrolytes, however, the mobility of lithium ions tends to be lower due to interaction with the donor atoms of the polymers. Thus, detailed investigations gave evidence for lower transference numbers of lithium ions in polymer electrolytes. In PEO-like polymer electrolytes the lithium ion transference numbers tend to be around 0.2-0.3 (32). There are also reports of cation transference numbers of 0.3-0.5, and with the help of additives, in the order of 0.8 (40-41).

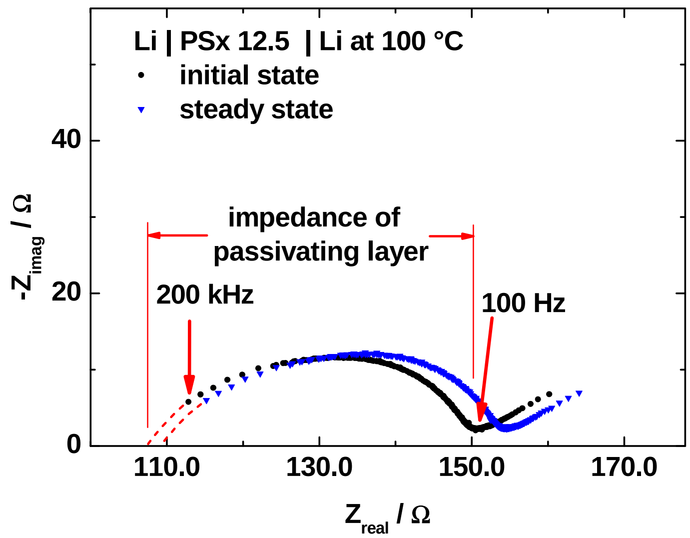

Figure 6: Complex plane plot of PSx 12.5 containing 0.61 mmol/$g_{polymer}$ LiBOB between two lithium foils (A = 1.13 cm²). The extrapolation on the left towards higher frequencies gives an estimate for the electrolyte resistance which agrees well with the results of impedance measurements using blocking electrodes.

Several techniques have been developed and published for polymer electrolytes in order to measure transference numbers (31, 42-46). If the transference number for lithium ions is very low, concentration polarization becomes dominant and can lead to salt precipitation causing serious problems in cell performance. The temperature dependence of the transference numbers is seldom measured, although it is also of interest to characterize the electrolytes at various temperatures. Wang *et al.* determined lithium transference numbers at 22 °C for a polysiloxane based polymer electrolyte with oligo ether side-chains connected with acrylate groups, no temperature dependent transference numbers could be found in the literature (17).

Fig. 6 shows the Nyquist plot of the impedance of one of our polymer electrolytes measured at 100 °C in the cell (Li|Polymer|Li). The diagram contains the data before applying a dc voltage and after reaching a steady-state polarization.

Fig. 7 shows the current vs. time during the potentiostatic polarization. A few minutes after switching-on the external dc voltage (10 mV), the current drops from 60 µA to a steady state current of 20 µA. The constant steady state current after about 40 min shows that the polymer system is stable at 100 °C in direct contact to elementary lithium. No spontaneous degradation or fluctuation of the heated cell was observed. The observed current relaxation was used to determine the transference numbers based on Eq. 1 and Eq. 2 (Table 1). To check the reproducibility of the measured values, two different sample membranes of the same salt-in-polymer material were analyzed for each temperature.

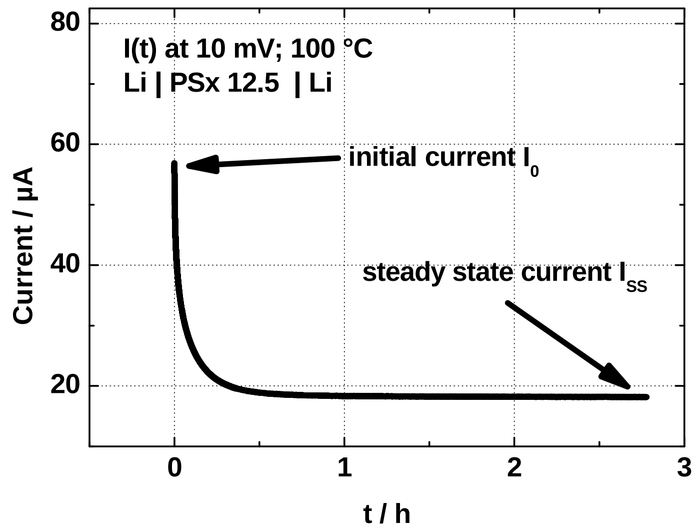

Figure 7: Potentiostatic polarization of a PSx 12.5 containing 0.61 mmol/$g_{polymer}$ LiBOB between two lithium foils (A = 1.13 cm²).

Fig. 8 shows the temperature dependence of the polymer electrolyte PSx 12.5 containing 0.61 mmol/$g_{polymer}$ LiBOB in a temperature range from 50 °C to 100 °C. With increasing temperature, the cation transference number increases from 0.18 to 0.28 (average values). The motion of the lithium ions is strongly coupled to the segmental motion of the polymer. A tentative explanation for the increasing transference number at higher temperatures may be explained by an increasing dissociation of the salt or/and a decrease of the association of lithium ions with the solvating oxygen donor atoms of the polymer, Of course, both, the cation and anion partial conductivities increase with temperature. (cf. Fig. 3). In total, the temperature dependence shows that the mobilities of both cations and anions start to approach each other at higher temperatures. The values of the transference number of lithium towards lower temperatures is well in line with the results of other groups on similar ether based polymer electrolytes while a temperature dependence has not been previously reported (17, 39).

TABLE I. Results of lithium transference number measurements at 70 and 100 °C.

I_0 / µA	I_{SS} / µA	R_0	R_{SS}	$R_{ELY.0}$	$R_{ELY,SS}$	eq. [1] t_{Li+}	eq. [2] t_{Li+}
70 °C							
22.0	7.0	226.8	227.5	171.4	177.1	0.19	0.20
23.2	7.4	209.1	210.9	164.1	167.1	0.20	0.20
100 °C							
57.3	19.3	45.2	66.3	91.7	95.0	0.29	0.30
56.9	18.2	35.5	47.3	102.3	116.8	0.28	0.32

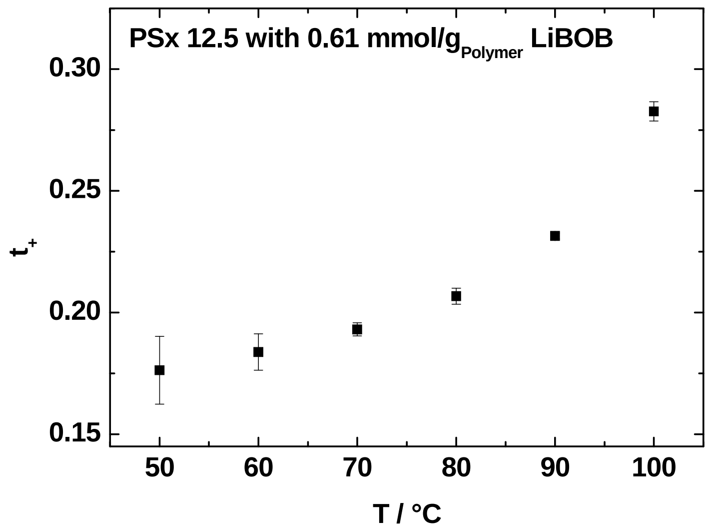

Figure 8: Lithium transference number measurements of PSx 12.5 containing 0.61 mmol/$g_{polymer}$ LiBOB at different temperatures calculated by eq. [1].

Conclusion

The stability limit vs. anodic oxidation of the polysiloxane based polymer electrolytes determined at platinum turned out to be excellent ranging between 4.5 V and 4.7 V (vs. Li/Li^+). Under cathodic conditions in the low potential range, very low current densities show that reduction of the polymer electrolyte components remains negligible until lithium plating takes place at -0.04 V vs. Li/Li^+. It is obvious that reduction of the polymer membrane is negligible. This, however, does not exclude the formation of a SEI on lithium metal as detected in the transference number measurements.

In conclusion, the results indicate that polysiloxane based electrolytes with stable boron derived salts can be used in a standard lithium ion battery up to elevated temperatures (70 – 100 °C), e.g. with conventional lithium iron phosphate as the cathode material. As no experiments were carried out with lithiated graphite as the anode material, we cannot predict the stability behavior for graphite at the moment.

Acknowledgments

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